Half-GCD candidate-root updates

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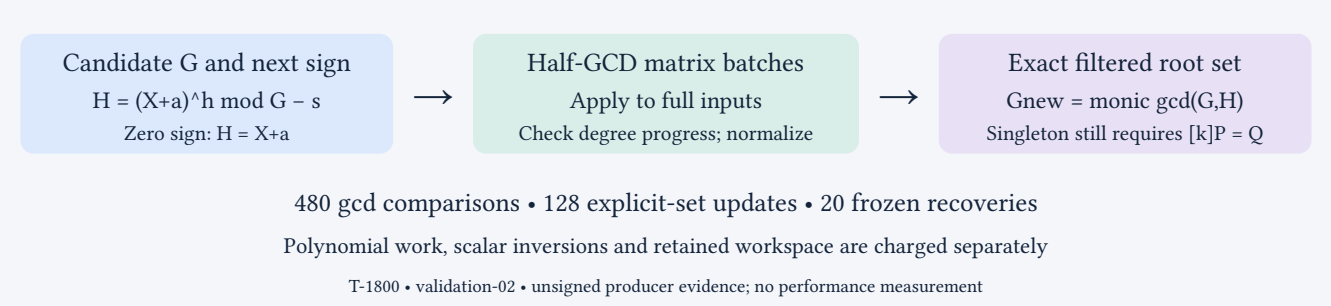
Candidate-root recovery now uses the existing recursive half-GCD engine for later sign constraints. A new capped driver returns a monic gcd and handles zero, constant, swapped and equal-degree inputs. The moment converter, seed construction, sign source and final scalar-to-point check are preserved.



Why the candidate set is preserved

Each Euclidean step is a polynomial matrix with determinant −1. Products of these matrices preserve the ideal generated by the input pair and therefore its gcd up to a unit. The existing half-GCD applies its matrix back to the full inputs and checks the degree reduction; final monic normalization fixes the representative.

For a squarefree split candidate polynomial, gcd with the next shifted character constraint retains precisely the matching roots. A zero sign uses X+a. The empty polynomial1 is absorbing. Recovery accepts a singleton only after checking its point equality.

Native checks

Validation-02 passes three selected methods,643 imported methods filtered. A separately coded schoolbook division/gcd gives480 exact comparisons across six primes, with planted common factors, zeros, equal and swapped inputs, sparse degree drops and degrees through130. These exercise224 outer half batches. Raw-length caps and modular coefficient reduction have controls.

Direct filtering of explicit candidate sets gives128 complete polynomial/count/first-moment update checks over F479 and F65537, starting with60 and256 candidates. They exercise235 recursive half calls. Repeated and contradictory signs, empty-state absorption, invalid offsets and invalid signs are checked.

All20 saved public-point recoveries on F509, $y^2=x^3+2x+10$, P=(3,68), r479 match the previous adapter in scalar, full transcript and initial degree, exercising242 recursive half calls. Third endpoint477, wrong candidate, coordinate, query and degree controls pass. These counters show path coverage, not performance.

Cost and remaining state

The standard gcd bound is O(M(n) log n) field operations. Charging the native exponentiation-based scalar inversions gives the conservative modular-multiplication bound O(M(n) log n + n log r). The sign-constraint power still costs O(M(n) log r), and source evaluation remains separate. Installed multiplication is Karatsuba; FFT constants and practical crossover thresholds are not claimed.

Root coefficients remain linear in n. Recursive matrices, transition metadata and polynomial temporaries are charged, with conservative O(n log n) workspace rather than a measured RSS figure. Larger-transcript aggregation without full root materialization remains the next mathematical task. No timing panel or cloud work ran; the original4,563-case cohort is unchanged.

Custody and sources

Thirty-seven source/input hashes are bound; generation pins the unchanged arithmetic and narrowly changed dispatch. Validation-01 stopped before compilation because four test includes retained old paths; exact source/validator snapshots and failure record remain. Validation-02 corrects only those paths. Test/normal warning lines68/87, including summaries, are retained.

Read existing half-GCD invariants and van der Hoeven, Optimizing the half-gcd algorithm (2022), introduction and arithmetic model. DERIVATION.md contains the source link and proof/accounting scope. KB unavailable. Report inspected; no independent signed verdict or measured speed comparison.